

Reservoir Seepage Quantification Using GRACE and Satellite Altimetry

Reservoir seepage is difficult to quantify in transboundary or data-scarce basins where ground monitoring is limited or politically constrained. This study presents a satellite-based framework that combines GRACE/GRACE-FO terrestrial water storage anomalies with satellite altimetry to isolate and estimate seepage in large reservoirs, addressing key gaps in the GRACE record through machine-learning-based reconstruction of missing months, and is validated against an independent GRACE-only benchmark with a mean amplitude error of under 0.5 inches. By separating surface reservoir storage, derived independently from an altimetry-based Area-Elevation-Volume curve ($R^2 = 0.994$, $RMSE = 1.87$ BCM), from total basin water storage, the framework isolates the subsurface storage signal and, combined with inflow and evaporation estimates, produces a seepage estimate derived entirely from satellite data, without reliance on in-situ leakage monitoring. We demonstrate the approach on the Grand Ethiopian Renaissance Dam (GERD), a rapidly filling reservoir central to an ongoing transboundary dispute over the Nile, where continuous and independently verifiable storage records are especially valuable. Preliminary results are consistent with GERD's known filling behavior. The framework is designed to generalize to other large reservoirs, offering a transferable tool for satellite-based reservoir monitoring in basins with limited ground infrastructure.